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TIDAL POWER GENERATION SYSTEM.

Abstract

A tidal energy capture system for powering a power generator by capturing tidal energy in a body of water experiencing tides, comprising: a base mounted to a ground; a float floating in the body of water adjacent the base; and an arm extending between the base and the float and pivotable relative to the base, the arm being connectable to the power generator for powering the power generator when the float is moved up and down relative to the base by the tides.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to the general field of power generation, and, more particularly, to a tidal power generation system.

BACKGROUND

[0002] Over the last century, a relatively large number of sea shore based and off shore based configurations of tidal power generation systems have been proposed to harness the renewable energy represented by the movements of the sea due to the tides. While these known tidal power generation systems of the prior art can generally fulfill the main objective of providing a means to convert this vertical movement of the sea tide into electrical or mechanical power, they also generally include one or more of the following disadvantages. For example, sea shore based tidal power generation generally require to be built in immediate proximity to the sea shore, if not partially into the waters near the sea shore. This location requirement often implies extended construction engineering expenses when the sea shore is represented by unstable grounds or is blocked by large and uneven rock formations.

[0003] Furthermore, the relatively complex functional configurations of sea shore based, as well as offshore based, tidal power generation systems of the prior art are generally not particularly well suited to sustain repetitive structural stress caused by the more and more frequent occurrence of category 4 or 5 sea hurricanes due to climate change.

[0004] Thus, there is a need for improved tidal power generation systems. An object of the present invention is to provide such improved systems.

SUMMARY OF THE INVENTION

[0005] In a broad aspect, there is provided a tidal energy capture system for powering a power generator by capturing tidal energy in a body of water experiencing tides, comprising: a base mounted to a ground; a float floating in the body of water adjacent the base; and an arm extending between the base and the float and pivotable relative to the base, the arm being connectable to the power generator for powering the power generator when the float is moved up and down relative to the base by tides.

[0006] There may also be provided a tidal energy capture system wherein the ground is above sea level coastal ground.

[0007] There may also be provided a tidal energy capture system wherein the ground is submerged ground located under the body of water.

[0008] There may also be provided a tidal energy capture system wherein the base is supported on the ground.

[0009] There may also be provided a tidal energy capture system wherein the base is buoyant and anchored to the ground.

[0010] There may also be provided a tidal energy capture system wherein the arm is telescopic.

[0011] There may also be provided a tidal energy capture system wherein the float is provided with a ballast system for selectively adjusting a buoyancy of the float.

[0012] There may also be provided a tidal energy capture system wherein the float is selectively submersible.

[0013] There may also be provided a tidal energy capture system wherein the ballast system includes a snorkel for accessing the atmosphere to pump air in the ballast system when emptying the ballast system.

[0014] There may also be provided a tidal energy capture system further comprising a hydraulic cylinder extending between the base and the arm and operative for selectively locking the arm relative to the base.

[0015] There may also be provided a tidal energy capture system further comprising underwater turbines mounted to a turbine mount, the turbine mount being mounted to one of the float and base.

[0016] There may also be provided a tidal energy capture system wherein the underwater turbines are movable between a retracted position and a deployed position, wherein in the retracted position, the underwater turbines are withdrawn inside the turbine mount, and, in the deployed position, the underwater turbines protrude from the turbine mount.

[0017] There may also be provided a tidal energy capture system wherein the underwater turbines are pivotable so that a rotation axis of the underwater turbines is movable between a substantially horizontal orientation and a substantially vertical orientation.

[0018] There may also be provided a tidal energy capture system wherein the turbine mount is at a side surface of the one of the float and base.

[0019] There may also be provided a tidal energy capture system wherein the turbine mount is at a bottom surface of the one of the float and base.

[0020] There may also be provided a tidal energy capture system wherein the underwater turbines include blades, the blades being fenced.

[0021] There may also be provided a tidal energy capture system wherein the turbine mount extends downwardly from the one of the float and base.

[0022] There may also be provided a tidal energy capture system wherein the turbine mount is rotatable about a vertical turbine mount rotation axis relative to the one of the float and base.

[0023] In another broad aspect, there is provided a tidal power generation system for capturing tidal energy in a body of water experiencing tides, comprising: a base mounted to a ground; a float floating in the body of water adjacent the base; an arm extending between the base and the float and pivotable relative to the base; and a power generator operatively coupled to the arm for providing power when the arm pivots relative to the base.

[0024] There may also be provided a tidal energy capture system wherein the power generator is an electric generator usable to provide electricity, the tidal power generation system further comprising batteries for storing the electricity, a shipping dock for allowing a ship to dock to the tidal power generation system, and a transshipment crane usable to move the batteries between the tidal power generation system and the ship.

[0025] Advantageously, the proposed power generation system, is relatively simple, and therefore likely to be economical and relatively easy to manufacture. Also, in embodiments in which the float is submersible, the float can be submerged if severe weather occurs, which minimizes the stress on the mechanical structures of the power generation system. In embodiments in which the underwater turbines are provided, tidal energy can also be captures in more than one manner.

[0026] The present patent application claims priority from UK Request for a Patent GB2403219.5 filed Mar. 5, 2024, the contents of which is hereby incorporated by reference in its entirety.

[0027] Other objects, advantages and features of the present invention will become more apparent upon reading of the following non-restrictive description of some embodiments thereof, given by way of example only with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. 1, in a perspective view, illustrates an embodiment of a tidal power generation system, according to the present invention, here shown comprising a stationary base rigidly anchored to the ground along a sea shore, and a float and arm arrangement extending from the stationary base, the float being shown in a first, or high tide, position;

[0029] FIG. 2, in a perspective view, illustrates the tidal power generation system shown in FIG. 1, here showing the float in a second, or low tide, position;

[0030] FIG. 3, in a perspective view, illustrates an other embodiment of a tidal power generation system, according to the present invention, here shown comprising a stationary base rigidly anchored to the sea floor and extending upwardly at least slightly above the sea surface at its highest sea tide level;

[0031] FIG. 4, in a side elevational view, illustrates yet another embodiment of a tidal power generation system, according to the present invention, here shown comprising a stationary base rigidly anchored in the sea floor and extending upwardly at an intermediary vertical position between the sea floor and the sea surface at its lowest tide level;

[0032] FIG. 5, in a side elevational view, illustrates the tidal power generation system shown in FIG. 4, here showing the float and arm arrangements submerged so as to avoid being damaged by a powerful sea hurricane raging along the sea surface;

[0033] FIG. 6, in a side elevational, cross-sectional view, illustrates an exemplary embodiment of a hydraulic cylinder in a close hydraulic circuit that is used as an actuatable braking means for selectively maintaining a predetermined angle of a float and arm arrangement relative to a stationary base, as exemplified in FIGS. 4 and 5;

[0034] FIG. 7, in a side elevational view, illustrates a telescopic boom usable in the tidal power generation systems of FIGS. 1 to 5;

[0035] FIG. 8, in a perspective view, illustrates underwater turbines mounted to a turbine mount, the underwater turbines being shown in a retracted position;

[0036] FIG. 9, in a perspective view, illustrates the underwater turbines shown in FIG. 8 in a partially deployed position;

[0037] FIG. 10, in a perspective view, illustrates the underwater turbines shown in FIG. 8 in a deployed position;

[0038] FIG. 11, in a perspective view, illustrates the underwater turbines shown in FIG. 8 mounted to a pivotable turbine mount;

[0039] FIG. 12, in a perspective view, illustrates the underwater turbines shown in FIG. 8 mounted to a float;

[0040] FIG. 13, in a partial perspective view, illustrates the underwater turbines shown in FIG. 8 mounted to a base tethered to the sea floor;

[0041] FIG. 14, in a perspective view, illustrates the underwater turbines shown in FIG. 8 mounted to the base;

[0042] FIG. 15, in a side elevation view, illustrates the underwater turbines shown in FIG. 8 mounted to the base;

[0043] FIG. 16, in a side elevation view, illustrates an underwater turbine protected by a net or grid; and

[0044] FIG. 17, in a side elevation view, illustrates an array of underwater turbines protected by a net or grid.

DETAILED DESCRIPTION

[0045] FIGS. 1 to 5 illustrate various embodiments, according to the present invention, of a tidal power generation systems **100**, **100'** and **100''**. In one embodiment of the tidal power generation system **100**, **100'** and **100''**, the latter comprises a stationary base **102**, at least one float and arm arrangement **104**, and at least one power generator **101**.

[0046] The stationary base **102** includes a stationary base lower end portion **106** mounted to a ground, for example by being stably anchored in an underlying ground surface, and a stationary base upper end portion **108** extending substantially upwardly relative to the stationary base lower end portion **106**. For the tidal power generation system **100**, the underlying ground surface is a dry land **300** location adjacent the sea shore **304**, or in other words an above sea level coastal ground, as seen in FIGS. 1 and 2 for example. For the tidal power generation system **100'**, the underlying ground surface is an off shore seabed **302**, or in other words ground located under a body of water receiving the float of the float and arm arrangement **104**, as seen in FIG. 3 for example. In this

embodiment, the stationary base upper end portion **108** is located a location that is at least slightly above the highest sea tide level relative to the seabed **302**. For the tidal power generation system **100'**, the underlying ground surface is also an off shore seabed **302**, but with the stationary base upper end portion **108** submerged and located at an intermediate location between the seabed **302** and the lowest sea tide level, as seen in FIGS. **4** and **5**.

[0047] Each float and arm arrangement **104** includes a float, for example a floating barge **110**, and an arm, for example taking the form of an elongated boom **112**. However, other types of floats and arms are usable in alternative embodiments of the invention, as long as a floating object can transmit the up and down movements experienced due to tides to the power generator **101** through the arm. The floating barge **110** is suitably sized and shaped for substantially stably floating on a sea surface experiencing tidal sea level variations. The floating barge **110** floats on the sea adjacent the stationary base **102** and is connected to the latter through the boom **112**.

[0048] As seen in FIG. **1** for example, the boom **112** has a substantially elongated configuration defined by a boom proximal end **113** pivotably connected to the stationary base **102**, a boom distal end **115** pivotably connected to the floating barge **110**, and an elongated boom intermediate portion **117** extending longitudinally between the boom proximal and distal ends **113** and **115**. Each float and arm arrangement **104** is suitably sized and configured so as to extend substantially laterally away from a respective peripheral portion of the stationary base **102**.

[0049] Furthermore, each float and arm arrangement **104** is pivotably movable relative to the stationary base **102** so as to be movable relative thereto between a first position and a second position, shown respectively in FIGS. **1** and **2**. In the first position, the floating barge **110** is floating on the sea surface when the latter is at its highest tide level. In the second position, the floating barge **110** is floating on the sea surface when the latter is at its lowest tide level.

[0050] Each power generator **101** is suitably sized and configured for converting a pivoting movement of the boom **112** relative to the stationary base **102**, into one of a usable electrical, mechanical or hydraulic power, among other possibilities. The power generator **101** therefore includes mechanisms and devices that convert the pivotal movement of the boom **112** to usable power. In a specific embodiment of the invention, the power generator **101** is an electric power generator.

[0051] The power generators **101** may for example be coupled between the stationary base upper end portion **108** and a respective one of the boom proximal end **113** pivotably connected thereto, between each boom distal end **115** and its respective floating barge **110**, or between the stationary base upper end portion **108** and simultaneously with the boom proximal end of all the float and arm arrangements **104**.

[0052] With stationary base lower end portion **106** anchored in the seabed **302** and the stationary base upper end portion **108** extending at least slightly above the highest sea tide level, as in the tidal power generation system **100'**, each boom distal end **115** is pivotably mounted to a top or side portion of the respective floating barge **110**. Furthermore, with stationary base lower end portion **106** anchored in the seabed **302** and the stationary base upper end portion **108** located at least slightly below the lowest sea tide level, as in the tidal power generation system **100''**, each of the boom distal ends **115** is pivotably mounted to a side or bottom portion of the respective floating barge **110**.

[0053] The power generators **101** may include transmissions or other power conversion devices that can take the relatively high torque and low speed movements of the booms **112** and convert these movements to usable power in conventional manners. Such power conversion devices are well-known in the art of tidal power generation and therefore not described in details herein.

[0054] In some embodiments, the tidal power generation system **100** may be advantageously used as a land electrical power station for a nearby village or sea port or as a stand alone off shore charging station for large and small boats and sea ships equipped with hybrid or fully electrically powered engines.

[0055] In some off shore embodiments of the invention, the stationary base **102** is a buoyant stationary base **102** that is rigidly and stably anchored in the seabed **302** using a suitable arrangement of anchor cables. A buoyant stationary base **102** thus anchored to the seabed may be used in the off shore embodiment of the present invention having a partially submerged stationary base **102**, as illustrated in FIG. 3, as well as the embodiment including a fully submerged stationary base **102**, as illustrated in FIGS. 4 and 5. Off shore buoyant stationary bases with anchor cable arrangements are well known in the art of, for example, off shore wind turbines, off shore oil well platform, and the likes.

[0056] In some off shore embodiments of the invention, as exemplified in FIGS. 3 to 5, the floating barge **110** is suitably configured, sized and equipped to serve as a floating charging station for large and small fishing boats, cargo ships and tourists cruise vessels equipped with hybrid or fully electrically powered engines.

[0057] For example, as exemplified in the figures, the floating barge **110** may be suitably sized and configured to include one or more shipping docks **120**, multiple battery storage space in its hold, and a transshipment crane **122** usable for transshipment of depleted shipping container size batteries **400** that are replaced with freshly charged ones on cargo ships **500** as illustrated. Thus, in a similar manner as truck stops distributed along highways, the off shore embodiments of the tidal power generation system **100** of the present invention may be advantageously installed at regular intervals along relatively long shipping routes.

[0058] In other embodiments, the transshipment crane **122**, or any equivalent actuable structure, may be also used to extend one or more charging cables to a ship located in proximity with the floating barge **110**. Furthermore, other known types and configurations of means for transshipment of batteries **400** between the floating barge **110** and the ships can be used.

[0059] In some off shore embodiments of the invention, as exemplified in FIGS. 3 to 5, the tidal power generation system **100'** or **100''** comprises a plurality of float and arm arrangements **104** that extend radially from the stationary base **102** and that are circumferentially equidistant from each other, so as to impart greater lateral stability to the tidal power generation system **100'** or **100''** during the cyclic vertical tidal movements of the sea. Thus advantageously, as best illustrated in FIG. 3, the radial distribution of the plurality of float and arm arrangements **104** allows simultaneous access to relatively large cargo ships **500** having hybrid or fully electrically powered engines to recharge or exchange depleted batteries **400** with freshly recharged ones.

[0060] In some off shore embodiments of the invention, as exemplified in FIGS. 4 and 5, the floating barge **110** further includes a suitably configured and sized ballast system **130** for selectively adjusting a buoyancy of the floating barge **110**, so that the floating barge **110** may be selectively temporarily sunk below the sea surface before a forecasted hurricane, and then refloated when the environmental conditions are appropriate. Thus, the most vulnerable portions of the tidal power generating system **100''** may be spared from potentially damaging weather conditions.

[0061] In some embodiments, the ballast system **130** is provided with a selectively retractable floating snorkel tube **132** for allowing access to surface air during refloating operations of the floating barge **110**, when the ballast system **130** is emptied of water, rather than using a system with compressed air tanks, which consumes significantly more energy to operate.

[0062] Alternatively, in some embodiments of the invention (not shown in the figures), surface air is provided to each floating barge **110** equipped with a ballast system **130** through a suitable centralized air conduit network extending to the surface via the partially submerged stationary base **102**, as illustrated in FIG. 3, or a centralized retractable snorkel via the fully submerged stationary base **102**.

[0063] Furthermore, in some embodiments of the invention, each one of the float and arm arrangement **104** further includes a hydraulic cylinder arrangement **140** operatively coupled between the stationary base **102** and the elongated boom **112**. The hydraulic cylinder arrangement **140** is one of a known closed circuit type via an actuable valve **142**, as exemplified in FIG. 6, or a

hydraulically powered type via a suitable hydraulic pump system (not shown in the figures). The hydraulic cylinder arrangement **140** may be used to stabilize the respective floating barge **110** during active charging or transshipment operations with one or more ships by locking the boom **112** relative to the base **102**. This operation can be effected by stopping the flow of hydraulic fluid within the hydraulic cylinder arrangement **140**.

[0064] It is also contemplated that the hydraulic cylinder arrangement **140** may be used as a selectively actuable brake system to maintain the respective floating barge **110** temporarily above its floating line once the surrounding sea surface has reached its highest sea tide level. Hence, the braking effect of hydraulic cylinder arrangement **140** may then be released so as to benefit from a greater gravity force pushing down on the elongated boom **112** and, thus, significantly increase the mechanical power available for conversion into usable electrical power.

[0065] In some embodiments of the invention, as seen in FIG. 7, the boom **112** is telescopic, as indicated by arrow **119**, and movable between a boom retracted position and a boom extended position, wherein the elongated boom **112** has a shorter length dimension in the boom retracted position than in the boom extended position.

[0066] The elongated boom **112** may be telescopically movable using any suitable telescopic actuator, such as an electrically or hydraulically powered actuator, for large applications of the tidal power generation system **100**, **100'** and **100''**. Alternatively, the elongated boom **112** may be user selectively and telescopically adjusted by hands in relatively small applications of the tidal power generation system **100**, **100'** and **100''**.

[0067] Advantageously, the tidal power generation system **100**, **100'** and **100''** thus provided with a telescopically movable elongated boom **112** may be relatively easily and economically adapted to various situations and environmental constraints using one and same construction or assembly configuration of the invention. For example, the elongated boom **112** may be telescopically adjusted so as to allow the stationary base **102** to be anchored on an economically advantageous location on dry land that is more or less near the sea shore, so as to have the booms **112** extend through a crevice between two elevated rock formations in proximity to the sea shore. Another advantage of the telescopically movable booms **112** resides in that the latter may be relatively easily length adjusted so as to be adapted to the specific transmission ratio of a new generator **101** installed in the tidal power generation system **100**, in replacement of a defective one.

[0068] Another advantage of the telescopically movable boom **112** resides in that, when a significant hurricane is forecasted in the region, the telescopically movable boom **112** may be extended so as to allow the floating barge **110** to be temporarily sunk below the sea surface further away from the sea shore where the seabed is relatively deeper and the surface waves less violent. Thus, the most vulnerable portions of the invention may be spared from potentially damaging weather conditions.

[0069] Referring collectively to FIGS. 8 to 17, in some embodiments, the tidal power generation system **100**, **100'** and **100''** may further comprise underwater turbines **200** mounted to a turbine mount **202**. The turbine mount **202** is mounted to the floating barge **110**, as illustrated in FIG. 12, to the base **102** if the latter is at sea, or to both the base **102** and floating barge **110**.

[0070] The underwater turbines **200** include blades **204** that can rotate when water flows through the blades **204**, for example due to tidal water movements or to any other water movement. The underwater turbines **200** include a conventional generator able to convert the rotational movement of the blades **204** to electricity using magnet and coil combinations. In some embodiments, as shown in the drawings, the blades **204** are surrounded by a shroud **206**.

[0071] The underwater turbines **200** are movable relative to the turbine mount **202** between a retracted position, shown in FIG. 8 for example, and a deployed position, shown in FIG. 10 for example. FIG. 9 illustrates a partially deployed position of the underwater turbines **200**. In the retracted position, the underwater turbines **200** are withdrawn inside the turbine mount **202**, so that the blades **204** don't protrude therefrom. In the deployed position, the underwater turbines **200**

protrude from the turbine mount **202**, so that any movement of water adjacent the turbine mount **202** can rotate the blades **204**.

[0072] In a specific embodiment of the invention, the turbine mount **202** defines recesses **208** each for receiving one of the underwater turbine **200** thereinto in the retracted position. The underwater turbines **202** are pivotally mounted to the turbine mount **202** so as to be movable between the retracted and deployed positions, for example through as motorized axle mounted to the turbine mount **202** (not shown in the drawings). In some embodiments, the range of motion between the retracted and deployed positions exceeds 90 degrees, for example about 135 degrees, which allows adjustment of the orientation of the underwater turbines **200** to maximize energy capture.

[0073] The turbine mounts **202** may be at a side surface **230** of the floating barge **110** or base **102**, at an underside surface **232** of the floating barge **110** or base **102**, or at an intersection between a side surface **230** and an underside surface **232** of the floating barge **110** or base **102**, as seen in FIG. **12** for the floating barge **110**. For example, the turbine mounts **202** are integrally formed in the floating barges **110**, so that in the retracted position, the underwater turbines **202** don't protrude from the floating barge **110**. The retracted position in such embodiments allows one to stow the underwater turbines **200** when needed, for example for maintenance of the floating barge **110**, when one wants to approach the floating barge **110** with a ship, or for any other reason.

[0074] In other embodiments, some or all the turbine mounts **202** protrude downwardly from floating barges **110** or from the base **102**, as seen in FIGS. **14** and **15**. It should be noted that in some embodiments provided with the underwater turbines **200**, the floating barges **110** and the float and arm arrangements **104** may be omitted, as tidal power is only captured by the underwater turbines **200**. In such embodiments, the base **102** may be anchored to the sea floor **302** through anchoring cables **210**. These anchoring cables **212** can be lengthened or shortened using a corresponding winch **214** mounted to the base **102**. In such embodiments, the anchoring cables **212** are not necessarily under a large tension, and may be used to maintain the base **102** around a suitable location, while allowing small horizontal drifts.

[0075] In such embodiments, as seen in FIG. **11** for example, the underwater turbines **200** may be pivotable so that a rotation axis **201** of the underwater turbines is movable between a substantially horizontal orientation and a substantially vertical orientation. This rotation allows one to maximize energy capture according to the direction of water flow during the tides. For example, shallower underwater turbines may have their rotation axis **201** horizontal, while deeper underwater turbines may have their rotation axis **201** vertical. To further maximize energy capture, the turbine mount **202** may also be rotatable about a vertical turbine mount rotation axis **205** relative to the base **102**.

[0076] In some embodiments, the blades **204** are fenced. For example, a net or solid grid **216** or **216'** is provided across the shroud **206**, as seen in FIG. **16**, or surrounding the entirety of the turbine mounts **202**, as seen in FIG. **17**.

[0077] To further energy production, the tidal power generation systems **100**, **100'** and **100''** may be provided with solar panels, wind turbines or small nuclear power generators, among other possibilities.

[0078] Although the present invention has been described hereinabove by way of exemplary embodiments thereof, it will be readily appreciated that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of this invention. Accordingly, the scope of the claims should not be limited by the exemplary embodiments, but should be given the broadest interpretation consistent with the description as a whole. The present invention can thus be modified without departing from the spirit and nature of the subject invention as defined in the appended claims.

Claims

- 1.** A tidal energy capture system for powering a power generator by capturing tidal energy in a body of water experiencing tides, comprising: a base mounted to a ground; a float floating in the body of water adjacent the base; and an arm extending between the base and the float and pivotable relative to the base, the arm being connectable to the power generator for powering the power generator when the float is moved up and down relative to the base by tides.
- 2.** The tidal energy capture system as defined in claim 1, wherein the ground is above sea level coastal ground.
- 3.** The tidal energy capture system as defined in claim 1, wherein the ground is submerged ground located under the body of water.
- 4.** The tidal energy capture system as defined in claim 3, wherein the base is supported on the ground.
- 5.** The tidal energy capture system as defined in claim 3, wherein the base is buoyant and anchored to the ground.
- 6.** The tidal energy capture system as defined in claim 1, wherein the arm is telescopic.
- 7.** The tidal energy capture system as defined in claim 1, wherein the float is provided with a ballast system for selectively adjusting a buoyancy of the float.
- 8.** The tidal energy capture system as defined in claim 7, wherein the float is selectively submersible.
- 9.** The tidal energy capture system as defined in claim 7, wherein the ballast system includes a snorkel for accessing the atmosphere to pump air in the ballast system when emptying the ballast system.
- 10.** The tidal energy capture system as defined in claim 1, further comprising a hydraulic cylinder extending between the base and the arm and operative for selectively locking the arm relative to the base.
- 11.** The tidal energy capture system as defined in claim 1, further comprising underwater turbines mounted to a turbine mount, the turbine mount being mounted to one of the float and base.
- 12.** The tidal energy capture system as defined in claim 11, wherein the underwater turbines are movable between a retracted position and a deployed position, wherein in the retracted position, the underwater turbines are withdrawn inside the turbine mount, and, in the deployed position, the underwater turbines protrude from the turbine mount.
- 13.** The tidal energy capture system as defined in claim 11, wherein the underwater turbines are pivotable so that a rotation axis of the underwater turbines is movable between a substantially horizontal orientation and a substantially vertical orientation.
- 14.** The tidal energy capture system as defined in claim 12, wherein the turbine mount is at a side surface of the one of the float and base.
- 15.** The tidal energy capture system as defined in claim 12, wherein the turbine mount is at a bottom surface of the one of the float and base.
- 16.** The tidal energy capture system as defined in claim 11, wherein the underwater turbines include blades, the blades being fenced.
- 17.** The tidal energy capture system as defined in claim 11, wherein the turbine mount extends downwardly from the one of the float and base.
- 18.** The tidal energy capture system as defined in claim 17, wherein the turbine mount is rotatable about a vertical turbine mount rotation axis relative to the one of the float and base.
- 19.** A tidal power generation system for capturing tidal energy in a body of water experiencing tides, comprising: a base mounted to a ground; a float floating in the body of water adjacent the base; an arm extending between the base and the float and pivotable relative to the base; and a power generator operatively coupled to the arm for providing power when the arm pivots relative to the base.
- 20.** The tidal power generation system as defined in claim 19, wherein the power generator is an

electric generator usable to provide electricity, the tidal power generation system further comprising batteries for storing the electricity, a shipping dock for allowing a ship to dock to the tidal power generation system, and a transshipment crane usable to move the batteries between the tidal power generation system and the ship.
